

Supplementary Figure S1. Hyperparameter search boundaries that yielded the best-performing configurations shown in Figure 4 of the main thesis

Random Forest hyperparameter boundaries

GridSearchCV

```
'n_estimators': [100, 200, 300]
'max_depth': [None, 10, 20, 30]
'min_samples_split': [10]
'min_samples_leaf': [1, 2, 4]
'criterion': ['gini', 'entropy']
```

Optuna

```
'n_estimators', [300, 400, 500, 600]
'max_depth', [None, 30, 40, 50]
'min_samples_split', [2, 5, 10]
'min_samples_leaf', [1, 2]
'criterion', ['gini', 'entropy']
```

XGBoost hyperparameter boundary

GridSearchCV and Optuna shares the same boundary

```
'learning_rate': [0.15, 0.2, 0.25]
'max_depth': [4, 5, 6, 7]
'n_estimators': [80, 100, 120, 150]
'colsample_bytree': [0.9, 1.0]
'subsample': [0.7, 0.8, 0.9]
```

CATBoost hyperparameter boundary

GridSearchCV and Optuna share the same boundary

```
'bootstrap_type': ['Bernoulli']
'subsample': [0.8, 1.0]
'iterations': [300, 600]
'depth': [4, 6, 8]
'learning_rate': [0.01, 0.1]
'l2_leaf_reg': [1, 3, 5]
'rsm': [0.8, 1.0]
```

SVM hyperparameter boundary

GridSearchCV and Optuna share the same boundary

```
'C': [0.1, 1, 10, 100]
'kernel': [linear, rbf, poly]
'gamma': [scale, auto]
'degree': [3]
```

Supplementary Table S1. Best-performing hyperparameter configurations for four classifiers (Random Forest, XGBoost, CatBoost, and SVM), corresponding to results shown in Figure 4.

Random Forest	n_estimators	max_depth	min_samples_split	min_samples_leaf	criterion
GridsearchCV (Full)	300	none	10	1	gini
GridsearchCV (patient-only)	100	none	10	4	gini
Optuna (Full)	400	30.0	10	1	gini
Optuna (Patient-only)	400	30.0	10	2	entropy

XGBoost	learning_rate	max_depth	n_estimators	colsample_bytree	subsample
GridsearchCV (Full)	0.15	7	80	1.0	0.8
GridsearchCV (patient-only)	0.2	7	80	1.0	0.9
Optuna (Full)	0.15	5	150	1.0	0.8
Optuna (Patient-only)	0.15	6	100	1.0	0.7

Catboost	bootstrap_type	depth	iterations	l2_leaf_reg	learning_rate	rsm
GridsearchCV (Full)	Bernoulli	4	300	1	0.01	1.0
GridsearchCV (patient-only)	Bernoulli	4	300	3	0.01	1.0
Optuna (Full)	Bernoulli	4	600	3	0.01	1.0
Optuna (Patient-only)	Bernoulli	4	600	3	0.01	1.0

SVM	C	degree	gamma	kernel
GridsearchCV (Full)	10.0	3	scale	rbf
GridsearchCV (patient-only)	0.1	3	auto	linear
Optuna (Full)	10.0	3	scale	rbf
Optuna (Patient-only)	0.1	3	auto	linear